Level: Bachelor
Programme: BE
Course: Software Engineering (New)

POKHARA UNIVERSITY

Semester: Fall

Year : 2025
Full Marks : 100
Pass Marks : 45
Time : 3 hrs.

Candidates are required to give their answers in their own words as far as practicable.
The figures in the margin indicate full marks.
Attempt all the questions.

1. a) Explain object-oriented analysis and its importance in requirements engineering. 7
   b) Compare and Contrast Waterfall and Incremental Model in detail. 8

   OR

   Explain Agile software development in detail. Compare Scrum and Extreme Programming with workflows and roles.
2. a) Explain software reuse approaches including frameworks, product lines, and COTS reuse with benefits and risks. 7
   b) Define risk. How do you identify risk in software engineering and also explain RMMM briefly. 8
3. a) Why is software engineering required? Describe the 4P's of software project management. 7
   b) Explain the Layered, client server and pipe and filter architecture with suitable diagrams. 8
4. a) What do you mean by black box and white box testing? Discuss the basis path testing and Cyclomatic Complexity of white box testing. 7
   b) What is software quality control and software quality assurance? Explain why reviews and inspections is essential in the SQA process. 8
5. a) Define software configuration management. Explain the process of system building and system release with suitable example. 8
   b) Why do we need to follow software process model while developing software projects. Justify above statement by taking one agile software process model highlighting its important aspects. 7
6. a) A customer goes to the restaurant and orders two pizzas and one coffee through restaurant application. Each table in the restaurant has LED display with the application installed. The customer must 8

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login to the restaurant application in order to place the order. After 20 minutes of placing the order food is served by a Robot at the custome's table. The customer pays the bills through the application and the system also allows the customer to provide feedback.

From this scenario

a. **Draw use case diagram**
b. **Sequence diagram**
c. **Class diagram**

b) For which type of software projects is the Spiral Model most appropriate? Justify your answer. 7

OR

What do you mean by agile development? Differentiate between Scrum and Extreme Programming. 2×5

7. Write short notes on: (**Any two**)

a) Alpha Vs Beta testing
b) Software Re-Engineering
c) Aspect oriented software engineering

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